GENERAL SPECIFICATION G7.12

Electric Traction Elevators

CONTENTS

G7.12 ELECTRIC TRACTION ELEVATORS

G7.12.1 Scope

(a) This Section of the Specification contains requirements which, where relevant to this Contract shall apply to Electric Traction Service Elevators.

G7.12.2 Standards and Codes

(a) Work included in the Section shall conform to the following, except as specified otherwise within this Section: ASME A17.1 - Safety Code for Elevators and Escalators.

(b) The works are to comply with the relevant requirements of the Standards and Codes listed below or referenced in the specification. Materials or products manufactured in accordance with alternative Standards and Codes may be proposed provided it can be demonstrated that the materials or products comply with the requirements of the specified standards. All Standards and Codes quoted are current version, unless specific year references are noted.

(c) The relevant codes include, but are not limited to the following as applicable to the equipment specified:

(i) SS CP 5;

(ii) SS CP 550;

(iii) SS 209;

(iv) SS 550;

(v) Code on Barrier-Free Accessibility in Buildings (BFA);

(vi) Code of Practice for Fire Precautions in Buildings;

(vii) Safety Code for Elevators and Escalators, ASME A17.1;

(viii) Fire Code 2013 Master Version;

(ix) Inspectors Manual for Elevators and Escalators, ASME A17.2;

(x) National Electrical Code, ASME/NFPA 70 and IEEE C1;

(xi) Life Safety Code, ASME/NFPA 101; and

(xii) Codes, ordinances, and laws applicable within the governing jurisdiction.

G7.12.3 Definitions

(a) Terms used are defined in the latest edition of the Safety Code for Elevators and Escalators, ASME A17.1.

(b) Reference to a device or a part of the equipment applies to the number of devices or parts required to complete the installation.

G7.12.4 Local Authorities

(a) Ensure that the works comply with the relevant authority regulations and directions.

(b) Where the terms ”testing authority”, ”approving authority”, “relevant authority’’, “local authority” or “local authorities” are used in this section, the bodies referred to are as follows:

(i) Building & Construction Authority (BCA);

(ii) Energy Market Authority (EMA);

(iii) Fire Safety & Shelter Bureau;

(iv) SPRING Singapore; and

(v) National Environmental Agency (NEA).

G7.12.5 Design Criteria

(a) Performance

(i) Speed: plus or minus (-) 5 percent of contract speed under any loading condition or travel direction;

(ii) Capacity: Safely lower, stop, and hold up to 125 percent of rated load;

(iii) Stopping Accuracy: plus (+) or minus (-) maximum 5mm under any loading condition;

(iv) Floor-to-Floor Performance Time: maximum seconds from start of doors closing until doors are three-quarters open, car level, and moving between floors; and

(v) Elevator data, such as car sizes, travel distance, door sizes, pit depth and headroom heights are as indicated on the Drawings.

(b) Standby Power Provisions

(i) Standby power of the same voltage characteristics via normal electrical feeders shall run elevator as designated in the Elevator Data Sheet on Drawings;

(ii) Conductors to signal power source from the standby power transfer switch to each elevator control panel in each elevator bank designated on the Elevator Data Sheet; and

(iii) Standby single-phase power to group controller and each designated elevator controller for lighting, exhaust blower, and emergency call bell.

(c) Fire-rated: 2-hour fire-rated requirement to the elevator landing door assemblies, PSB labelled.

(d) Green Feature: BCA Green Mark for Non-Residential Building Version NRB: 2015.

(e) Minimum 600mm pit clearance height space after elevators landing to level 1 per BCA circular.

(f) Do not use elevator for construction purposes or during building construction period without written permission from the Contractor or S.O..

G7.12.6 Quality Assurance

(a) Elevator manufacturer shall have a service office located in Singapore. The office shall stock a full complement of parts and supplies necessary for servicing specified equipment.

(b) Perform welding in accordance with AWS D1.1.

(c) Pre-installation Conference

(i) Hold a preinstallation conference at the project site 2 weeks prior to commencing work. All persons directly involved with the work of this Section shall attend; and

(ii) Review schedule of installation, installation procedures and conditions, and coordination with related work.

(d) Perform tests required by consultant, governing authority, and in the presence of authorized representatives. Supply personnel and equipment for final tests and inspections within the Contract.

(e) Coordination

(i) Before submitting product data and shop drawings, coordinate location of elevator equipment, rails and devices connecting elevator components to building construction with mechanical systems, electrical services and controls, and building construction.

(ii) Provide coordination drawings indicating no conflicts or obstructions.

(f) Delivery, Storage, And Handling

(i) Deliver materials in manufacturer's original, unopened protective packaging;

(ii) Store material in original protective packaging. Prevent soiling, physical damage, and wetting. Store in a dry, warm place as directed by the S.O.; and

(iii) Protect equipment and exposed finishes during transportation, erection, and construction against damage and stains.

(g) Submittals

(i) Provide the following:

1) Standard Product Data on the following items:

- Load and dimensions for both cabs and shafts;
- Signal and operating fixtures, operating panels, and indicators;
- Cab design and components;
- Door and frame materials and details, including on cabs and shaft walls;
- Electronic equipment to control and monitor elevator control functions; and
- Pit and shaft headspace height requirements, machine room height requirements.

(ii) Provide the following shop drawings within 2 weeks after Contract award:

1) Scaled and dimensioned layouts of cabs and shafts;

2) Plans of pit, hoist way, and machine room indicating equipment arrangement;

3) Elevation section of hoist way, details of shaft wall door, frame and sill;

4) Provide coordination drawings showing arrangement of elevator equipment in relation to adjacent building construction and other equipment;

5) Button panel opening design on shaft wall and machine room maintenance hook drawing;

6) Control panel drawings with time settings; and

7) Cable route drawings for supervisory panel.

(iii) Provide the following within 6 weeks after Contract award:

1) Cab interior elevations and ceiling layouts;

2) Details of car enclosures, door and frames;

3) Remote / Supervisory panel drawings;

4) Machine room equipment layouts;

5) Design Information: Indicate equipment lists, reactions, and design information on layouts, including rail bracket spacing and reactions;

6) Power Confirmation Sheets: Include motor horsepower, code letter, starting current, full-load running current, and demand factor for applicable motors;

7) Elevator recall control diagrams;

8) Expected heat release and starts per hour, and minimum and maximum equipment room operating temperature requirements for elevator equipment in machine room;

9) Finish Material: Submit 300 mm by 300 mm samples or 300 mm lengths of actual finished materials for S.O. review of colour, pattern, and texture only. Compliance with other requirements is the responsibility of the elevator contractor. Include signal units, push buttons, lights, graphics, and mounting methods;

10) Fixtures: cuts, samples, or shop drawings;

11) At elevator control panel, identify contacts for primary and alternate recall; and

12) Show contact point for emergency power shutdown.

(iv) Operation and maintenance data.

(h) Project Record Documents

(i) Provide reproducible drawings of wiring and control diagrams and interior car details.

(i) Operation and Maintenance Data

(i) Provide, in addition to the requirements of General Specification G1.2 Information Management. Include the following at a minimum:

1) Straight-line wiring diagram of as-installed elevator circuits with index of location and function of components. Mount installation wiring diagrams on panels, racked or similarly protected, in elevator machine room. Maintain with addition of all subsequent changes. These diagrams are S.O.'s property.

2) Lubricating instructions, including recommended grade of lubricants. Mount on copy behind plastic or nonglare glass glazing in metal frame mounted on machine room wall.

3) Parts catalogues for all replaceable parts, including ordering forms and instructions.

4) Provide cabinet for parts storage.

(ii) Keys: Deliver three sets of keys for elevator controls and doorway keyways to Contractor.

(j) Maintenance Service

(i) Include with New Equipment Contract (Warranty Period):

1) Comprehensive preventive maintenance and 24-hour emergency call-back service with response time to site for main fault within 2 hours and lift passenger trap to attend within 30 mins on all equipment described herein for a period of 24 months commencing on date of Substantial Completion. Perform work in accordance with the signed comprehensive preventive maintenance agreement. Systematically examine, adjust, clean, and lubricate all equipment. Repair and replace defective electrical and mechanical parts using parts produced by the manufacturer of installed equipment. Maintain elevator machine rooms, hoist ways, and pits in clean condition.

2) Perform maintenance work without removing cars from service during peak traffic periods.

3) Use competent personnel supervised and employed by elevator contractor.

4) The S.O. may delete cost of maintenance from new equipment contract and remit 24 equal instalments directly to the elevator contractor during period in which work is being accomplished.

(ii) Contract (Ongoing Preventive Maintenance Program): Quote monthly cost for 5-year comprehensive and non-comprehensive maintenance agreement commencing on completion of the 24-month period in paragraph A above. Based on current costs; price adjustment will be allowed at commencement date and thereafter as provided in agreement.

G7.12.7 Products

(a) General

(i) Traction elevators and elevator machine rooms located within the building shall be as shown on the Drawings and as described on the elevator data sheets. The elevators shall be interconnected with the emergency power system and with the building smoke and fire alarm system.

(ii) Fabricate and assemble various parts in shop to minimize field assembly. Assemble parts which require close field fit in shop and mark for field erection.

(b) Materials

(i) Steel

1) Sheet Steel (Furniture Steel for Exposed Work): stretcher-levelled, cold-rolled, commercial-quality carbon steel complying with ASTM A366, matte finish;

2) Sheet Steel (for Unexposed Work): hot-rolled, commercial-quality carbon steel, pickled and oiled, complying with ASTM A569; and

3) Structural Steel Shapes and Plates: ASTM A6, ASTM A36, and ASTM A108.

(ii) Stainless Steel: SS 304; No. 4 finish - brushed.

(iii) Aluminium: ASTM B221, extruded 6063 alloys with T6 temper; A212 finish.

(iv) Painting

1) Structural Metal Surfaces: Clean surfaces of rust, oil, or grease; wipe clean with solvent; prime two coats with rust-inhibitive zinc-molybdate, alkyd primer (FS TT-P-645), 0.1 mm dry film thickness.

2) Machine Room Components: Clean and degrease; one coat rust-inhibitive alkyd primer, 0.05 mm dry film thickness; and two coats alkyd semi-gloss enamel, 0.1 mm dry film thickness.

3) Galvanized Surfaces: Clean with neutralizing solvent in conformance with SP-1 surface preparation specifications of Steel Structures Painting Council. Prime one coat with zinc-dust zinc-oxide primer (FS TT-P-641), 0.025 mm dry film thickness.

(v) Primer for Plain Steel Surfaces: FS TT-P-645, primer, paint, zinc-molybdate, alkyd type.

(vi) Electrical Components

1) Conduit: 19 mm minimum size, EMT;

2) Fittings: steel setscrews or steel compression;

3) Termination: Use compression termination on feeder and motor connections. Wire nut termination permitted on lighting and acceptable termination for circuits not exceeding 20A. Use terminal strips with separate pressure foot to compress conductor;

4) Conductors: 600V class, standard copper; XHHW insulation 4 AWG and larger, THHN/THWN No. 6 and smaller;

5) Support raceways and boxes from structure. Do not support electrical components from other system or bodies;

6) Point of Connection: Provide junction box or space within enclosure for termination of field conductors provided by others. Location of point of connection shall be readily accessible and labelled for the type of service to be connected, e.g., lighting, fire alarm, telephone, fire recall (primary and alternate), and power off;

7) Spare Conductors: Include 10 percent extra conductors and two pairs of shielded audio cables in traveling cables;

8) Do not parallel conductors to increase current-carrying capacity unless individually fused;

9) Do not use metal conduit as sole means for grounding conductor. Provide individual ground conductor in each raceway. Ground conductor shall be sized not less than NEC Table 250-95;

10) Protect elevator equipment against damage or malfunction due to change to or from normal power supply; and

11) Include wiring and connections to elevator devices remote from hoist way.

(c) Operation

(i) Selective Collective

1) Operate elevator without an attendant from buttons located at each floor and in car. When elevator is idle, automatically start car and dispatch it to floor corresponding to registered car or hall call. Once the car starts, respond to registered calls in direction of travel in the order floors are reached.

2) Do not reverse car direction until all car calls have been answered or until all hall calls ahead of car and corresponding to direction of car travel have been answered.

3) Slow down and stop car automatically at floors corresponding to registered calls in the order in which they are approached in each direction of travel. As slowdown is initiated for a hall call, automatically cancel the call and render the hall button for that direction of travel ineffective until the car leaves floor. Cancel car calls in same manner. Hold car at arrival floor for an adjustable time interval to allow passenger transfer.

4) Answer calls corresponding to direction in which car is traveling unless call in the opposite direction is highest (or lowest) call registered.

5) Illuminate appropriate button to indicate call registration. Extinguish light when call is answered.

(ii) Other Items

1) Load Weighing: Provide means for weighing passenger load. Design control system to provide dispatching in advance of normal intervals and to provide landing call bypass when the car is filled to adjustable percentage of rated capacity (adjustment range, 10 percent to 100 percent).

2) Anti-nuisance Feature: If car loading is not commensurate with registered car calls, cancel car calls.

(iii) Door Operation: Open doors automatically when the car arrives at a floor to permit transfer of passengers. Automatically close doors after a timed interval.

(iv) Automatic Stopping Accuracy: Stop car within 6 mm above or below the landing sill. Avoid overtravel, as well as under travel, and maintain stopping accuracy regardless of load in car, direction of travel, rope slippage, or stretch.

(v) Independent Service: Provide controls for operation of each elevator from car buttons only. Close doors by constant pressure on desired destination floor button. Open doors automatically upon arrival at selected floor.

(vi) Motion Control: dc, variable-voltage type with closed-loop feedback suitable for operation specified and capable of providing smooth, comfortable acceleration, retardation, and dynamic braking. Limit the difference in speed between full load and no load to not more than plus or minus 5 percent of the Contract speed. Design, install, and adjust elevator equipment to meet the performance requirements of paragraph 2.3 within the following parameters:

1) Horizontal Acceleration Within Cars During All Riding and Door Operating Conditions: not more than 25 mg peak to peak in the 1- to 10-Hz range;

2) Acceleration and Deceleration: constant and not more than 4.5 fps with an initial ramp between 0.5 and 0.75 second; and

3) Sustained Jerk: not more than 7 fps squared.

(vii) Firefighters' Service: per code to operate and recall elevators to designated or alternate designated floors in fire or other emergency condition. Recall shall occur via a Form C dry-contact relay provided by the fire alarm system. Relay shall be located within the elevator equipment room. Provide sensor signal wiring from machine room connection point to controller terminals. Operate visual/audible signal until return is complete or automatic operation restored.

(viii) Standby Lighting and Alarm: car-mounted battery unit with solid-state charger to operate alarm bell and lighting per code. Battery to be rechargeable with 5-year minimum life expectancy. Provide test button in service cabinet of car station which causes illumination of standby lighting bulbs. Arrange system so lights are part of normal car lighting system.

(ix) Standby Power Transfer: If normal power fails, adequate standby power will be supplied through normal power feeders to start and run elevator at rated speed.

(x) Sleep Mode Feature: Elevator to go into sleep mode where elevator automatically switch off lights and ventilation fan when elevator idles for 5 minutes, to meet BCA Green Mark requirement.

(d) Lift Supervisory Control Centre

(i) The Lift Supervisory Control Centre shall be located at the Fire Command Centre. The Panel shall be fabricated from 1.6mm (16 SWG) sheet steel with a hinged 2mm (14SWG) hairline stainless steel faceplate, finished to the requirements of the S.O.. The panel shall include the following minimum displays and controls:

1) A digital read out type car position indicator as specified elsewhere in this Specification and a car-in-service LED indicator for each lift, together with directional arrows arranged to remain illuminates showing the direction of travel and the direction of intended travel at the next operation when the car is stationary at a landing.

2) The digital read out type car position indicators shall also indicate all the floors within the blind hoist way of each of the lifts concerned.

3) The car lights shall not remain illuminated when a lift is switched out of service. The indicating lights shall be in the form of letters or numerals as required. UP directional arrows shall be green and the DOWN directional arrows red.

(ii) LED lights to indicate the mode of operation of each lift viz:

1) Light indication for group/automatic operation;

2) Light indication when “Attendant” key is switched on; and

3) Light indication when “Independent Service” key is switched on.

(iii) A flashing light for each lift which shall be activated when any of the associated lift is jammed in the hoistway. The digital read out car position indicator of the lift concerned should indicate where the lift is jammed including the blind hoistway.

(iv) LED lights to indicate the source of power under which each group of the lift systems is operating

(v) LED lights to indicate if any of the lifts is on emergency due to either over speed, activation of the emergency alarm button, when the safety gear is applied or any other emergency to cause the shutdown of the lift. An alarm bell shall be provided in the panel which shall be set off during any of these emergencies. An alarm muting button shall be provided to cancel the audible alarm; however, the visual indication shall remain until the fault is cleared. A different colour light such as red shall be used.

(vi) LED lights to indicate if any of the lifts is taken off for maintenance.

(vii) LED lights to indicate if any of the lifts fails to start; lights shall be the flashing type.

(viii) A 'Car to Main Lobby' key switch for each and every lift, the operation of which shall cause the respective lift to bypass all car and hall calls and proceed to the main lift lobbies and park there with the car doors opened. After a pre-set time interval, the associated car lighting and ventilation fans shall be switched off. The lifts can only be made operational again by resetting the respective 'Car to Main Lobby' key switches. For the Fire Lifts, the operation of the respective "FIREMEN SWITCH" shall defeat the above-mentioned mode of operation.

(ix) The key switch shall be constructed in such a way that it cannot be removed unless it is in the "Automatic" position.

(x) A separate key switch shall be provided to reset the lifts after receipt of a fire alarm signal

(xi) All indicating lights, key switches, terminal blocks, etc. used shall be of the proprietary make, of adequate ratings and to the relevant Singapore Standards. Three sets of keys shall be provided for each type of key switch called for. The layout of the Lift Supervisory Control Panels shall be reviewed and approved by the S.O..

(e) Machine Room Equipment

(i) Arrange equipment in spaces shown in bid drawings. Provide identifying numbers on machine, power conversion unit, controller, and disconnect switch.

(ii) Gearless Traction Machine

1) Each gearless traction machine shall include a slow speed direct current motor, traction driving sheave and electro mechanical brake, all mounted on a steel bedplate. The bearing of the machines shall be of the roller type and shall be dust proof and provided with adequate means of lubrication. The brake pulley and traction sheave on the motor armature spindles shall be designed to minimise torsional stress in the main shaft. The hoisting motors shall be of the DC reversible type with high starting torque and low starting current, particularly designed for lift service to operate with contract load and speed without overheating. Provisions shall be made to enable the speed to be checked by a portable tachometer.

2) Brakes – Every lift machine shall be provided with a brake that is mechanically applied and electrically held off. The brake shall comprise at least two brake shoes with non-combustible linings and shall be capable of bringing the lift car to rest smoothly, under maximum conditions of load and speed. The brake shall also be capable of sustaining a static load of 125% of the contract load. No brake shall be released in normal operation unless electrical power is applied to the lift motor. No earth fault, short circuit or residual magnetism shall prevent the brake from being applied when the power supply to the lift motor is interrupted. Means of releasing the brakes in emergency shall be provided. Provisions should also be made to ensure immediate re application of the brake as soon as hand pressure is released.

3) Insulation & Construction - All hoisting machines shall be electrically insulated to Class ‘B’ of BS EN 60085. The construction of all lift machines shall conform generally with Clause 2.15 of BS 2655: Part 1.

4) Vibration Isolation - All hoisting machineries, including motors, transmissions, brakes and pullers shall be isolated from the supporting structure with minimum 12mm static deflection double-deflection neoprene mounts, sized (rebearing surface area) to accommodate the loads at the various support points. Any restraints or anchoring bolts, if provided, shall be isolated from the mounting base and/or structure with neoprene inserts of min. 10 mm thickness. The entire mounting system shall be totally captive and in event of failure of the pad mounts, shall not allow equipment/base to move more than the thickness of the pads.

(iii) Solid State Drive Units

1) A full solid state drive unit of appropriate design shall be provided for each DC gearless lift to effect voltage control by means of supplying a uniformly varying DC voltage to each lift motor.

2) Conversion from 3 phase AC voltage to variable DC voltage shall be accomplished by means of back to back power converters using thyristors or by means of two parallel bridges of three pairs of silicon controlled rectifiers (SCR) or other acceptable means. Close loop control system shall be employed to provide positive, smooth response at all speeds and loads.

3) The solid state drive units shall be suitable for operation on 400V +/- 10%, 50 Hz AC supply and shall each be rated to suit the running and accelerating conditions of the associated lift motor. The drive units shall be suitable for operation in ambient temperature up to 40 degree C and relative humidity of 100%.

4) Each drive unit shall comprise solid state plug in type modules, main contactors, reed switches, indicating lamps, filters, wiring and other necessary components all housed in a compact free standing steel cabinet with perforated doors complete with necessary ventilation fans. The construction of the cabinets shall be similar to that specified elsewhere for equipment panels.

5) Protective devices shall be provided for each drive unit to shut down the lifts concerned upon sensing of a phase reversal or single phasing.

6) Each drive unit shall be provided with the necessary isolating transformer, suppression and filtrational devices etc. to reduce the harmonic distortion to the levels specified in IEEE 519 as well as to reduce the noise to acceptable level.

(iv) Variable Voltage & Variable Frequency Drive System

1) The drive system shall be full solid state microprocessor-based comprising converters, inverters with digital regulators etc. to drive the respective AC induction traction motor using pulse width modulation (PWM) control.

2) The drive shall provide quiet and smooth operation with high degree of levelling accuracy, high efficiency (low power consumption) and good power factor under all load condition and direction of travel.

3) Isolation and/or filtration devices/circuits shall be provided to effect noise control and also to minimise the harmonic distortion to the AC power supply source such as the distortion is restricted to less than 5% of the operation current waveform.

(v) Controller - Individual Car and Group

1) Frame: Securely mount all assemblies, power supplies, chassis switches, relays, and other items on a substantial, self-supporting steel frame. Completely enclose equipment with covers and ventilate to prevent overheating.

2) Switch and Relay Design: direct-current type, magnet operated with contacts of design and material to ensure maximum conductivity, long life, and reliable operation without overheating or excessive wear and provide a wiping action to prevent sticking due to fusion. Provide switches carrying highly inductive currents with arc deflectors or suppressers.

3) Microprocessor-Related Hardware:

- Fabricate printed circuit boards with FR4 or G10 glass epoxy material with a minimum equivalent 1 ounce of copper;
- Coat all printed circuitry with tin lead;
- Include built-in noise suppression devices which provide a high level of noise immunity on double-sided printed circuit boards;
- Include built-in noise suppression devices which provide a high level of noise immunity on all solid-state hardware and devices;
- Provide power supplies with noise suppression devices;
- Isolate inputs from external devices (such as push buttons) with optoisolation modules;
- Provide separate regulated power supply for each computer chassis;
- Design control circuits so that one side of power supply is grounded to provide for testing;
- Under no circumstances shall the safety circuits be affected by accidental grounding of any part of the system;
- Design the system so it will start properly when power is restored in the event of a power failure or interruption;
- Provide system memory so data is retained in the event of power failure or disturbance;
- Design or protect equipment so it will operate without difficulty when exposed to 500-kHz to 106-Hz radio signal at a power of 100 watts within a distance of 3 meters; and

- Design or protect equipment to provide electromagnetic interference (EMI) shielding within FCC guidelines.

4) Power Supplies: UL or CSA recognized with short-circuit protection.

5) Wiring: CSA-labelled copper wires for factory wiring. Neatly route all wiring interconnections and securely attach wiring connections to studs or terminals.

6) Permanently mark components (relays, fuses, PC board, etc.) with symbols shown on Drawings.

7) Provide extender boards when computing devices are used inside a computer chassis to facilitate access to the printed circuit cards utilized.

8) Use stable capacitor or crystals as the time base for electronic time-delay devices.

(vi) Provide separate panel for interconnection of functions not driven by elevator components (emergency telephone, car speaker, or power shutdown, for example).

(vii) Sleeves and Guards: steel angle guards around cable or duct slots if overhead grating is required.

(viii) Machine and Equipment Support Beams: structural steel beams required for direct support of elevator machine, 2:1 sheaves, deflector sheaves, overhead sheaves, governor, and dead-end hitches. Provide bearing plates, anchors, shelf angles blocking, embedments, etc., to support machine beams or equipment.

(ix) Provide hold-down bolts for machines located beside the hoist way. Isolate machine and overhead sheave beams to eliminate noise and vibration transmission to building structure.

(x) Governor: centrifugal type, car driven, with pull-through jaws and bidirectional electrical shutdown switches. Provide overhead supports required for attachment to building structure.

(xi) Hoist Machine Drip Pans: metal containers to collect lubricant seepage.

(xii) Noise and Vibration Control

1) To minimize noise and vibration in occupied areas, mechanically isolate elevator equipment (including hoist machines, deflector sheaves, power conversion units, and support equipment) from the structure; electrically isolate controllers, machine motors, and power conversion units.

2) Limit noise level relating to elevator equipment and its operation to no more than:

- 50 dBA in elevator cars under any condition, including door operation and exhaust blower on highest speed; and
- 70 dB(A) measured at 1 meter away from the lift machinery and 1 meter above the floor in machine room.

(f) Hoist way Equipment

(i) Guide Rails: planed steel T sections suitable for elevator travel, car weight, and counterweight weight with brackets for attachment to building structure. Provide backing and intermediate counterweight tie brackets to meet code and seismic requirements.

(ii) Buffers: spring type with blocking and supports. Provide switch on buffers to limit elevator speed in down direction if buffer is compressed.

(iii) Sheaves: machined grooves with ball or roller bearings. Provide mounting means to machine beams, machine bed plate, car and counterweight structural members, etc.

(iv) Pit-Tensioning Sheaves: Mount sheaves and frames on pit support members or guide rails. Provide with guides or pivot points to enable free vertical movement and properly tension cables/tapes.

(v) Counterweight: steel frame with metal filler weights, guided by roller guide shoes.

(vi) Counterweight Guard: metal guard around counterweight in pit per elevator code.

(vii) Hoist and Governor Ropes

1) 8 by 19 or 8 by 25 construction, traction-steel type;

2) Governor rope to suit manufacturer's specification; and

3) Fasten with adjustable shackles/wedge sockets.

(viii) Normal and Final Terminal Stopping Devices: per elevator code.

(ix) Electrical Wiring and Wiring Connections

1) Provide two-way intercom between lift car lift machine room, fire command centre and lift supervisory panel. System shall consist of handset type master units, speaker/microphone type, slave units with interphone or alarm mute button module on car operating panel.

2) Conductors and Connections: copper throughout with individual wires coded and connections on identified studs or terminal blocks. Use no splices or similar connections in wiring, except at terminal blocks, control cabinets, junction boxes, or condulets. Provide 10 percent spare conductors throughout. Provide nine pairs of shielded communication wires in addition to those required to connect specified items. Run spare wires from car connection points to individual elevator controllers in the machine room. Tag spares so they can be identified in the machine room.

3) Conduit: galvanized steel or EMT conduit and duct. Do not use flexible conduit exceeding 36 inches in length. Flexible heavy-duty service cord may be used between fixed car wiring and car door switches for door protective devices.

4) Traveling Cables: flame- and moisture-resistant outer cover. Prevent traveling cables from rubbing or chafing against hoist way or elevator equipment within hoist way.

5) Life Safety: Connect smoke sensors, telephones, and speakers, etc., to interface panel in machine room.

(x) Entrance Equipment

1) Door Hangers: two-point suspension with upthrust adjustment. Tire rollers so that no metal-to-metal contact exists;

2) Door Tracks: bar or formed, cold-drawn steel with smooth hanger contact surface. Provide removable tracks or track surface for replacement;

3) Interlocks: Provide type operable without retiring cam; and

4) Closers: spring, counterweight, or spirator type.

(xi) Pit Stop Switch: per elevator code.

(xii) Floor Numbers: stencil-painted 100 mm-high floor numbers within the hoist way per elevator code.

(xiii) Access: hoist way access switches mounted in terminal floor entrance jambs and access keyways at each floor.

(g) Hallway Car Entrances

(i) Hallway door frames and doors shall be supplied and installed for all the lifts at all lift lobbies where called for. The door frames and angle steel brackets shall be suitably braced and reinforced.

(ii) Frames

1) Door frames shall be of bolted construction of a unit assembly comprising head and jamb sections made of steel of at least 2.0 mm thickness (14 SWG).

2) The frames shall be provided with adjustable wall anchors or comparable devices to permit bonding of these anchors or devices into the walls after the frames are in place.

3) All frames shall be securely fastened to sills and hanger supports, and shall be returned on the hoistway side to present a neat appearance.

4) The door shall be acoustically treated and fire rated as specified for the car doors. The finishes of the door frames, jambs, transom panels and doors of the various lifts at the lift lobbies are as shown in this Specification.

5) All hall doors together with the frames shall be of one-hour fire rating to the authorities' approval and each door shall bear a PSB's test label to certify the fire rating.

6) Both edges of the landing door at every floor landing shall be engraved with floor numberings of 75mm high.

(iii) Hanger supports shall be of sufficiently thick formed sections securely bolted to the steel strut angles, complete with integral hanger tracks.

(iv) Structural steel strut angles shall be formed of sufficient size to support the entrance and to accommodate the door closing equipment. The angles shall be continuous and securely bolted to the sills and the building structure. Where the hoistways are of dry wall construction, all necessary additional struts, beams, bracketing, etc. for receiving all the lift door hanger supports at all landings of all lift concerned to the building structure shall be deemed to have been included in the Sub-Contract.

(v) Hangers cover plates shall be made of best grade steel, removable and so arranged to assure hanger accessibility from within the lift car for maintenance purposes.

(vi) For all the lifts sheet steel fascia plates of thickness not less than 2.0 mm (14 SWG) shall be provided and fixed between the underside of landing entrance sills and the top of the door hanger case as well as between every blind hoistway emergency access doors to form a flush surface in the path of travel at the car entrance.

(vii) All hoistway emergency access doors shall also be provided with fascia plates. The fascia plates shall be installed for the full width of the clear hoistway door opening and shall be painted as detailed in Section GC of this Specification.

(viii) The clearance between the edge of the car-platform sill and the fascia plates including those installed at between the hoistway emergency access doors shall not be more than 125 mm as stipulated in the current edition of the Singapore Standard CP550: Code of Practice for Installation Operation and Maintenance of Electric Passenger and Goods Lifts.

(ix) Where there is more than one entrance to the lift car, the sides of the hoistway facing the other entrances shall also be provided with fascia plates.

(x) Material and Finish of Doors and Frames: stainless steel.

(h) Hall Call Buttons

(i) General

1) Hall call buttons shall be provided and installed as shown on the Specification Drawings.

2) The hall call button stations shall each comprise a hairline stainless steel faceplate to the S.O.'s requirements. The buttons shall be of circular or square shape and shall be so designed that it will illuminate and register a call when pressed. When a car arrives and cancels the call, illumination will cease. If a non-stopping car passes through any indicated floor, however, the button shall remain illuminated.

3) Hall call buttons shall be the electronic call registered micro-movement push buttons, flush mounted on the faceplate.

(ii) The following indication are required for each Lift:

1) On Lowest Terminal Floor (or Floors for Double Deck Lifts)

- An 'Up' button at each entrance.

2) On All Intermediate Floors

- ‘Up’ and ‘Down’ hall call functions shall be provided for the hall call stations at all the intermediate floors.

3) On Top Terminal Floor

- A ‘Down’ button at each entrance.

(i) Hall Car Position Indicators & Directional Lanterns

(i) General

Combined hall car position indicators and directional lanterns shall be provided at the landings for the respective lifts as called for in Schedule ‘B’. For the other floors, hall directional lanterns only shall be provided.

(ii) Construction

The combined hall car position indicator and directional lanterns shall be the horizontal type each comprising a hairline stainless steel faceplate, a green ‘Up’ arrow and/or a red ‘Down’ arrow and a digital read out type car position indicator as specified in this Specification. As soon as a car has reached a predetermined distance from a particular floor, the directional lantern indicating the direction of travel of the lift car shall be illuminated, independent of any operation of a hall call button, and shall remain illuminated until the car has left the landing. Should a car over travel a short distance, the hall lantern for that floor shall remain illuminated to indicate the car’s original direction of travel.

(iii) Directional Lanterns

Hall directional lanterns shall be of similar construction as the combined unit with ‘UP' arrow at the lowest terminal floor, a ‘DOWN’ arrow at the top most floor and the ‘UP’ and ‘DOWN’ arrows at all intermediate floors.

(iv) Arrival Chime

A two stroke arrival chime shall be provided at each hall landing for each car to be incorporated into the directional hall lantern or the combined unit of the floor concerned. The chime should give distinct two tone audible sound upon arrival of the car concerned at each landing.

(j) Dust Covers and Toe Guards

(i) Suitable dust cover and toe guards fabricated from 2.0 mm (14 SWG) sheet steel shall be provided at the appropriate landings within the hoistway of each lift.

(k) Car Frames

(i) Construction

1) Every car shall be carried in a steel frame sufficiently rigid to withstand the operation of the safety gear without permanent deformation to the car frame. The safety factor for these frames shall be of not less than five (5).

2) At least four sets of renewable roller guide shoes shall be provided for each lift, two at the upper edges and two at the lower edges of the car frame. These guides shall be held in contact with the guide rail surfaces by means of adjustable cushioning devices and they shall run to on dry, unlubricated guide rails.

3) Nylon sliding guide shoes may be used for lifts of speed not exceeding 105 mpm.

(l) Car Platforms

(i) Construction

1) Car platforms shall be of framed construction. The platform for each of the passenger lifts shall be designed on the basis of the contract load being evenly distributed. These platforms shall each consist of a structural steel frame, a substantial timber floor and floor finishes.

2) The structural steel frame of the car platforms shall be anti-rust treated and painted as specified elsewhere in this Specification. The timber floor shall be of thoroughly seasoned teak or other suitable wood and suitably treated to protect against the ingress of moisture and from the growth of fungus and termite attack. The underside of the timber flooring shall be fire proof by means of sheet steel coated with an appropriate fire proofing material.

(ii) Clear Platform Size

1) Clear platform area of the lift shall be related to the car capacity called for and in accordance with the current edition of the Singapore Standard CP550: Code of Practice for Installation, Operation and Maintenance of Electric Passenger and Goods Lifts.

2) For purpose of design, the minimum factors of safety shall be 5 for steel and 8 for timber.

(iii) Car Floor Finishes

1) The floor finishes for the lift cars shall be in accordance with Schedule 'B' of this Specification. Door sills shall be of extruded aluminium alloy section with a non-slip wearing surface as for the landing door sills.

(iv) Car Wall Finishes

1) The wall finishes of the lift cars shall be in accordance with this Specification.

2) Heavy duty protective padded canvasses suitable for the protection of the three (3) sides of the car walls when the lift is used for movement of goods shall be provided for selected lifts as specified in schedule of this Specification.

3) The protective canvasses shall be the hook-on type and hooks for mounting of the canvasses shall be so installed in the lift cars concerned that they are not exposed to sight when the canvasses are removed.

(v) Car Ceiling

1) Ceiling for all cars shall be of steel and shall be painted with anti-rust enamel with a top coat. Provisions shall be made for ventilation of the lift car as specified below. Car ceilings of lifts shall be finished in accordance with Specification.

2) The internal clear height of the car cage shall be as specified.

(vi) Car Handrails and Car Kick Plates

1) Car handrails where called for in the contract shall be mounted so that the top does not exceed 900 mm from the car floor in accordance with the Code for Barrier Free Accessibility. The finishes of the car kick plates for all the lifts shall be as specified.

(vii) Illumination

1) Illumination General

- The minimum average illuminance on the floor of all lift cars shall be 150 lux using fluorescent fittings or incandescent downlights to be located above the suspended ceiling of the lifts, or surface mounted, where appropriate to the S.O.'s requirements.

2) Emergency Lighting

- Emergency lighting shall be provided for all the lift cars in accordance with the Lift Code requirements.

(viii) Ventilation

1) General

- Each lift car shall be provided with adequate ventilation using one or more fans of appropriate capacity, to be located above the suspended ceiling or recessed in the car ceiling as appropriate, all subject to review and approval by the S.O..
- Low speed and low noise blower fans shall be employed such that the internal noise levels of each lift car shall not exceed NC 45 with the car doors closed.

2) Emergency Power Supplies

- Emergency power supplies shall be provided to the ventilation fans in accordance with the Code requirements.

(ix) Emergency Exits

1) General

- Emergency exits shall be provided for the lift cars. The exits shall be hinged in the roof of the lift car and panels for top openings shall not open inwards. The exits shall be clear of any apparatus mounted above the roof of the lift car and be held by suitable fasteners that can be opened from outside the lift car and not from inside.

2) Interlock

- Panels for all emergency exits including those on the ceiling and floors of the double deck lifts shall be provided with an interlocking electrical switch which will prevent operation of the lift when the panel is open.

(m) Car Position Indicator

(i) General

1) One horizontal electrically operated car position indicator of appropriate design shall be installed above each car entrance transom panel inside every lift car.

2) LED type car position indicator and call button at every floor landing c/w up and down indicator

(ii) Type

1) All car position indicators shall be the digital read out, alpha numeric type in green display of not less than 50 mm in height with over 160 degree angle of view for continuous floor location reading including those of the blind hoistways.

(iii) Stainless Steel Faceplate

1) Each car position indicator shall be flush mounted on a hairline stainless steel faceplate to the S.O.'s requirements.

(n) No Smoking Notice

(i) A 'No Smoking' notice with an appropriate logo and the following message shall be provided for each lift car; and

(ii) The sign shall be reversed silk screened on an acrylic panel or engraved on a stainless steel panel as required by the S.O..

(o) Capacity Plates

(i) The contract load of each lift in kilograms and in number of persons, where appropriate, shall be engraved onto each of the car operating panels, the sizes and design of the letterings shall be subject to review by the S.O..

(p) Door Safety Devices

(i) Electronic door detectors shall be installed on the leading edge at high and low levels of each car to protect passengers entering or leaving the car. The detector shall produce a wide angle electrostatic field of detection which shall move well ahead of the closing doors, and reverses them without physical contact. A late passenger shall be able to walk through the entrance while the doors are closing without causing unnecessary re opening. The timing devices of the door operators and/or control system shall be operated in conjunction with the door detectors to provide adjustable, reduced hold open time once the detectors are activated and the doors re opened.

(ii) In addition to the electronic door detectors, mechanical door safety edges of anodized aluminium in black or silver metallic extending a minimum of 45 mm beyond the leading edge of car doors for the full height of the entrance shall be provided for the car door of each lift; the safety edge on any of the car door panels, on touching the passenger, a compression of not more than 13 mm shall cause the door to reopen immediately to prevent crushing the passenger. The doors shall recluse immediately thereafter. When the doors are fully closed, the edges shall retract to position flush with door panels.

(q) In Car Camera

(i) An in car CCTV camera to be provided for safety viewing, and to be linked and monitor to be provided in Control Room viewing.

(r) Car Operating Panels

(i) Car operating panels shall be stainless steel finish and each panel shall contain the following:

1) An “ALARM” button which, when pressed, shall cause battery operated alarm bells to sound at the following locations:

- Lift Supervisory Control Centre;
- Outside the lift hoist way of the lift concerned at the designated main lift lobby; and
- Lift Supervisory Repeater.

2) All alarm bells, batteries and wiring required shall be provided under this Sub-Contract.

(s) Speech Synthesizer

(i) An English speech synthesizer c/w microprocessor, vocabulary chips, loudspeaker volume control etc. for broadcasting to the car passengers, the floors arrived, direction of car travel, messages for obstructed doors, lift full, power interruption, technical difficulties and other announcement in conjunction with above. The sequence and frequency of the announcement of all messages shall be to the S.O.’s requirements.

(ii) For each of the goods lifts, a warning buzzer which shall sound automatically whenever the door starts to close.

(iii) For each of the goods lifts, an indicating light and a push button which when pushed on will cause the door hold open time to be prolonged to a predetermined period of time (adjustable up to 5 minutes). The indicating light shall be illuminated to show that the door hold open time has been prolonged. Before the expiry of the prolonged door time, it shall be possible to deactivate (i.e. off) the push to enable the doors to close automatically. Upon expiry of the prolonged door time, the passenger shall also be able to activate the button again in which case the doors shall be held open again for the predetermined period.

G7.12.8 Execution

(a) Inspection

(i) Verify that hoist way, pit, and machine room are ready for work of this Section;

(ii) Verify shaft and openings are of correct size and within horizontal and vertical tolerances;

(iii) Confirm electrical power is available and of correct characteristics (arrange for temporary electrical power to be available for installation work and testing of elevator components, if necessary);

(iv) Report defects or deficiencies in writing; and

(v) Beginning of installation means acceptance of conditions.

(b) Installation

(i) Install each equipment item in accordance with manufacturer's directions, referenced codes, and specifications;

(ii) Install machine room equipment with clearances in accordance with referenced codes and specifications;

(iii) Install items so they may be easily removed for maintenance and repair;

(iv) Install items so that access for maintenance is safe and readily available;

(v) Install equipment to afford maximum safety and continuity of operation in the event of seismic activity; and

(vi) Coordinate installation of front hoist way walls with installation of hoist way entrances.

(c) Tolerances

(i) Align guide rails vertically with tolerance of 1.6 mm in 30 m. Secure joints without gaps and file any irregularities to a smooth surface.

(d) Adjusting and Cleaning

(i) Balance car to equalize pressure of guide shoe rollers on rails;

(ii) Lubricate all equipment in accordance with manufacturer's instructions;

(iii) Adjust motors, generators, brakes, controllers, levelling switches, limit switches, stopping switches, door operators, interlocks and safety devices to achieve required performance levels;

(iv) Keep work areas orderly and free from debris during progress of project. Remove packaging materials on a daily basis as equipment is installed;

(v) Remove all loose materials and filings resulting from work;

(vi) Clean machine room equipment and floor of dirt, oil, and grease; and

(vii) Completely remove protective covering from finished surfaces. Clean hoist ways, cars, car enclosures, entrances, operating and signal fixtures, and trim, removing dirt, oil, grease, and finger marks.

(e) Field Quality Control

(i) General: Furnish labour, materials, and equipment necessary for tests. Notify S.O. 14 days in advance when ready for final review of each elevator.

(ii) Work at the jobsite will be checked during the course of installation. Full cooperation with reviewing personnel is mandatory. Accomplish corrective work required prior to performing further installation.

(iii) Have code authority acceptance inspection performed and complete corrective work. Provide copy of completed inspection report to S.O..

(iv) Final Acceptance - Final acceptance of installation will be made only after all field quality control reviews have been completed, identified deficiencies have been corrected, all submittals and certificates have been received, and the following items have been completed to satisfaction of Lift Contractor appointed BCA registered authorised examiner, and code authority:

1) Workmanship and equipment comply with Specifications;

2) Contract speed, capacity, and floor-to-floor performance comply with Specifications; and

3) Performance of following are satisfactory:

- Starting, accelerating, running;
- Decelerating, stopping accuracy;
- Door operation and closing force;
- Equipment noise levels;
- Signal fixture utility; and
- Overall ride quality.

(v) Temperature Rise Test: temperature rise in windings limited to 50 degrees C above ambient. Conduct a full-capacity, 1-hour running test, stopping at each floor for 10 seconds in up and down directions if equipment performance is questionable in S.O.'s and code authority's opinion BCA.

(vi) Acceptance Test Results: In all test conditions, obtain specified speed, performance times, floor accuracy without releveling, and ride quality to satisfaction of Lift Specialist / Contractor Authorised Examiner and to be inspected and approved by BCA.

(f) Manufacturer's Services

(i) Warranty Period Operating Test - Perform the following tests in the presence of S.O.: At an agreed time during warranty period, conduct tests to verify performance. Measure individual elevator performance functions and verify compliance with specifications and manufacturer's operation and maintenance guide.

(ii) Performance Guarantee - Should tests reveal defects, poor workmanship, variance, or noncompliance with requirements of specified codes and/or ordinances or variance or noncompliance with the requirements of Specifications, complete corrective work to satisfaction of S.O. at no additional cost:

1) Replace equipment that does not meet code or Specification requirements;

2) Perform work and furnish labour, materials, and equipment necessary to meet specified operation and performance; and

3) Perform and assume cost for retesting required by governing code authority or S.O. to verify specified operation and/or performance.

(g) Maintenance

(i) To provide full maintenance service by skilled competent employees of the lift installer for a period of 12 months following the date of substantial completion.

(ii) The monthly preventive maintenance shall be carried out during normal office hours.

END OF SECTION